Enhancing Efficiency with DevOps Solutions for Hybrid Cloud Platforms

Cloud-based DevOps solutions are in demand in the age of AI, as they enable operational efficiency. We take a quick look at five popular tools in this domain, and also explore the connection between CloudOps and observability.

Large enterprises tend to prefer hybrid cloud solutions. They share some workloads in the public cloud and some of the legacy solutions in their on-premises data center (DC) or private cloud to address strategic goals like cost of operation, security, and regulatory restrictions. Sometimes they also prefer a multi-cloud and cloud agnostic solution. Here, the private cloud or on-premises DC is replaced with multiple cloud service provider (CSP) based solutions. For example, they may opt for database solutions in Google Cloud Platform (GCP) while the application remains in Microsoft Azure.

Best practices for cloud agnostic DevOps architecture

As per Gartner, most large enterprises adopt a cloud-first strategy for digital transformation. Cloud architects recommend cloud agnostic architecture in order to provide a CSP neutral solution. In such an architecture, CSP-specific services (which cannot be ported across CSPs) and standard reusable services (which can be ported across CSPs) have to be separated.

Some best practices to keep in mind when building cloud agnostic architecture are:

- Using containers for microservices (Docker or Kubernetes).
- Using cloud neutral solutions like Terraform templates for configuring container orchestration in GCP or EKS.
- Using automation as much as possible.
- Avoiding usage of direct vendor lock-in services like native storage.
- Using SpringBoot microservices in cloud storage in Azure, GCP or AWS.
- Automating configuration (infra as a code) to reduce CSP dependency.
- Integrating cloud native directory and authentication services (e.g., Cognito or Google Directory or Azure AD).